

Dr. M. Usman Maqbool Bhutta

POSTDOCTORAL ASSOCIATE · HORTICULTURAL SCIENCES DEPARTMENT, UNIVERSITY OF FLORIDA

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Robotics & AI Researcher

Summary

Dr. M. Usman Maqbool Bhutta is a robotics and AI researcher with over **9+ years of specialization in robotics R&D and AI**, currently serving as a Postdoctoral Associate at the University of Florida. He specializes in multi-agent robotics, phenomics, and spatio-temporal tracking. A former startup co-founder (**COO/CEO**), Dr. Bhutta holds a **Ph.D.** from **HKUST** (ranked top 20 globally) and has conducted research at **ETH Zurich**. His work includes five top-tier journal publications (**IF: 31.2**) and conference papers at **CVPR, ICRA, RAL, and IROS**, earning him the **Best Student Paper Finalist Award** at IEEE CYBER 2018.

Work Experience

Research (Computer Vision Technologies and Robotics System)

Fruit Phenomics Team (FPT), UF/IFAS Horticultural Sciences Department

UF, US.

Postdoctoral Associate

Feb. 2025 - Present

- Supervisor: **Prof. Patricio R. Muñoz**, Director of **Blueberry breeding and Genomics Lab**, University of Florida **UF**, United States.
- Collaborating with **Dr. Bruno Leme** on several fruits phenomics-related projects, focusing on innovative solutions for agricultural productivity.
- Research Projects**
 - Farm-ng Rover Development for Blueberry Fields:** Developed and improved farm-ng rover for autonomously data collection in blueberry fields, focusing on machine learning model enhancement and computer vision techniques.
 - Computer Vision Team Mentorship:** Mentored computer vision team in improving machine learning models for multimodal feature extraction, yield estimation, acoustic analysis, and color-based image segmentation.
 - 3D Plant Reconstruction** Collaborated on 3D reconstruction of plants in the field for robot simulation and yield estimation improvement.
 - Rabid Breeding Chamber Project** Assisted engineering team in the Rabid Breeding Chamber project, contributing to the overall goal of establishing a robust phenomics team to support growers and companies nationwide.

Computer Vision and Sensing Systems (COVISS), Agricultural and Biological Engineering

UF, US.

Postdoctoral Associate

Aug. 2023 - Feb. 2025

- Supervisor: **Prof. Henry Medeiros**, Director of Computer Vision and Sensing Systems (**COVISS**) Lab, University of Florida **UF**, United States.
- Research Projects**
 - Long-term Visual Place Recognition:** Worked on a graph-based image feature representation to address the dynamic nature of regions. Utilized semantic information for enhanced place recognition.
 - Calibration-free Multi-view Detection and Association:** Focused on the importance of occlusion in improving MODA and MODP in multi-view object detection. Implemented attention networks to enable calibration-free optimization of multi-view detection. This work has been accepted in CVPR 2025.
 - End-to-End Detection, Segmentation, and Tracking:** Aiming to surpass state-of-the-art (SOTA) methods for the MOTs challenge through this project.
- Co-supervised the ABE Robotics team participating in the ASABE Robotics Student Design Competition, 2024, in Anaheim, California.
- Managed project documentation and code versioning, with all projects being trained on the UF HiPerGator.

C3 Robotics Laboratory (C3RL), Department of Mechanical and Automation Engineering

CUHK, Hong Kong.

Postdoctoral Research Fellow

May 2021 - Jun. 2023

- Supervisor: **Prof. Darwin Lau**, Director of C3 Robotics Laboratory (**C3RL**), The Chinese University of Hong Kong **CUHK**, Hong Kong.
- Research** in place recognition, AR/VR and inspection robotics for construction
 - Place Recognition:** Working on the light-weight representation of images and making better use of previously trained models.
 - AR/VR:** Research in **AR/VR** Platform design, multi-agent mixed reality, assistive prosthetics for the handicapped persons, teleoperated robots, human-robot interaction.
 - Inspection:** Deep learning based research for the robotic presurface inspection for the construction.
- Multi-agent collaboration:** For indoor vegetation and painting robotics application.

Autonomous Systems Lab (ASL)

ETH Zürich, Switzerland.

Academic Guest

Oct. 2019 - Mar. 2020

- Supervisor: **Prof. Roland Siegwart**, Director of Autonomous Systems Lab (**ASL**), ETH Zürich, Switzerland.
- Co-Supervisor: **Dr. Cesar Dario Cadena Lerma**, Senior Researcher at ASL, ETH Zürich, Switzerland.
- Research in Place Recognition:** I've worked for making the better place recognition for the better scene understanding. Furthermore, I've also introduced light weight representation of key-frames for better channel utilization.
- Development using:** MATLAB, Python, Jupyter, Nvidia CUDA, Docker, Deep learning, Matconvnet

Robotics and Multi-Perception Lab, Robotics Institute

HKUST, HK

PhD

Sep 2016 - Feb 2021

- PhD Supervisor: [Prof Ming Liu](#), Director of Robotics and Multi-Perception Lab ([RAM-LAB](#))

Research Contributions as First Author

Place Recognition: Our work makes a leap towards hierarchical deep learning-based place recognition for multiagent SLAM systems. We introduce a probabilistic layer to constrain the local representation along with the global representation of images. The global description of the pre-trained model and the local consistency introduced in our work enables the system to yield good performance at large-scale.

Development Using: MATLAB, Python, Jupyter, Nvidia CUDA, Docker, Deep learning, Matconvnet

Multi-Agent SLAM: I've produced Loop-box for the multi-Agent Direct SLAM Triggered by Single Loop Closure for Large scale semi dense 3D reconstruction /large-scale structure from motion. This work is accepted for publication in IEEE Transactions on Cybernetics (IF= 11+)..

Development using: C++ 11, ROS, OpenCV, OpenGV, Paraview, EVO, UGV, LSD-SLAM, Monocular Camera.

SLAM: I've worked mainly for the multi-agent SLAM system. To find the scale difference and excellent relative transformation between difference agents. I've produced [PCR-Pro](#) work. This work got best student paper finalist award at IEEE-CYBER 2018.

Development using: C++ 11, ROS, OpenCV, OpenGV, Paraview, EVO, Monocular Camera, Plotly

Automation (Inspection): For smartphone glass inspection, I've introduced [Smart-Inspect](#) work. This work is accepted in IROS 2020. The patent is also under-review in the US.

Development using: Python, Jupyter, Nvidia CUDA, K-Means, Docker, Machine learning (ML)

Department of Electronics

QAU, Pakistan

MPhil

Mar. 2010 - Jun. 2012

- M.Phil. Supervisor: [Prof Hasan Mahmood](#), Department of Electronics, QAU.
- Research in Image Processing (CNN):** I've introduced CNN-based license plate recognition of Pakistani's Cars. This work has been published in IEEE-ICALIP 2014.

Industry and Entrepreneurship

Lisee Technology Corporation Limited

Central, HK

COO

Jan 2018 - Jan 2021

- Lisee helps in the precise localization of (10 cm) and navigation in the indoor environment.
- Our Lisee's project demo can view in this [video](#).
- Won funding support for company | TSSSU Support HK\$ 300,000 (Jun 19 - May 20).
- Indoor Navigation for Libray using Lisee Technology | PCF Support HK\$ 160,000 (Jun 18 - May 19).

Friends Corporate Technologies

Islamabad, PK

CEO

Feb 2013 - Aug 2015

50+ Projects | Electronics/ Electrical Engineering, R&D, image processing, Web Applications Development, MVC Applications.

Federal Government of Pakistan

Islamabad, PK

Scientific Officer

May 2012 - Feb 2013

Electronics/ Electrical Engineering, R&D, image processing.

Server4Sale, LLC

Islamabad, PK

PHP Developer

Dec 2009 - Mar 2010

Php web application development for Joomla and Zend frameworks.

Academic (Teaching)

Department of Computer Science

BZU-Sahiwal, Pakistan

Lecturer (Visiting)

Feb 2016 - Jun 2016

Teaching MCS and BS-CS courses

Department of Physics

GPGC Sahiwal, Pakistan

CTI

Oct. 2015 - May. 2016

Teaching BS physics (Electronics related courses + Labs + Projects)

Department of Electronics

QAU, Pakistan

Instructor

Aug. 2010 - Dec. 2010

'PHP Web Development' (a short course) offered in Department of Electronics, Quaid-i-Azam University, Islamabad, Pakistan.

Education

Ph.D. in Robotics and AI at [RAM-LAB, Robotics Institute](#)

HKUST, Hong Kong

Department of Electronic & Computer Engineering, The Hong Kong University of Science and Technology

Sep 2016 - Jun 2021

- **Thesis:** “Towards A Swift Multiagent SLAM System For Large-scale Robotics Applications”, June 17, 2021
- PhD Supervisor: [Prof Ming Liu](#), Director of Robotics and Multi-Perception Lab ([RAM-LAB](#))
- Exchange Supervisor: [Prof. Roland Siegwart](#), Director of Autonomous Systems Lab ([ASL](#)), [ETH Zürich](#), Switzerland.
- Exchange Co-Supervisor: [Dr. Cesar Dario Cadena Lerma](#), Senior Researcher at ASL, [ETH Zürich](#), Switzerland.
- Won PEEF CCMS PhD scholarship from Pakistan and overseas research award (ORA).
- Research: Multi-Agent SLAM System, Place Recognition, Autonomous Car and Deep Learning for automation industry. Hand on experience on RGB-D 3D reconstruction, Multi-view stereo, python and C++ development experience, multi-threaded coding, 3D mesh registration, paraview and parameterizations. Moreover, I’ve deep understanding of data structures and algorithms. Experience with software development on Linux open-source stack. Calculus, linear algebra and geometry related courses have been taken.

M.Phil. in Electronics

QAU, Pakistan

Department of Electronics, Quaid-i-Azam University, Islamabad, Pakistan.

Mar 2010 - Jun 2012

- **Thesis:** “A Feature Based Algorithm for Automobile License Plate Detection and Recognition in Pakistan”, Mar, 2012
- M.Phil. Supervisor: [Prof Hasan Mahmood](#), Department of Electronics, QAU.
- Research: Signal and Image Processing

M.Sc. in Electronics

QAU, Pakistan

Department of Electronics, Quaid-i-Azam University, Islamabad, Pakistan.

Sep 2007 - Jun 2010

- **2nd position** in batch
- Projects: completed 9+ projects related to C++, Verilog/HDL and Micro-controller programming.

B.Sc. in Mathematics and Physics

BZU, Pakistan

Bahauddin Zakariya University, Multan Pakistan

Sep. 2005 - Aug. 2007

- Grade: 1st Division

Presentation

Navigating Academia Research & Industry, ITCN Asia 2023

*Pak-China Friendship Centre,
Islamabad*

Panelist

Feb. 2023

- ‘I emphasize taking the voluntary initiative by the tech industry first. Companies should realize that they need academic involvement to grow at a swift pace. Most VPs of big companies belong to academia.
- ‘In return, companies would at least solve their quality hiring problems. Usually, universities need more funding for R&D and IP filings, etc. In a broader view, both could get huge benefits.

Teaching and Learning Innovation Expo 2021

CUHK, Hong Kong

Posters Presentation

Jul. 2021

- ‘Mixed Reality in Hands-On Learning of Robotics’ by Xinyan Zhao, Sabrina Lam, M. Usman M. Bhutta, Darwin Lau and Jimmy Lee.
- ‘Online Robotics Laboratory Framework for Interactive and Group Hands-On Learning’ by Ken Hui, Ken Kwok, M. Usman M. Bhutta, Darwin Lau and Jimmy Lee.

IEEE/RSJ International Conference on Intelligent Robots and Systems [IROS, 2020]

Las Vegas, USA

Presenter as author

Oct. 2020

- Introduced **Smart-Inspect** for the micro scale localization and classification of smartphone glass defects for industrial automation.

Technology Start-up Support Scheme for Universities (TSSSU)

HKUST, Hong Kong

Presenter as COO Lisee Technology Corporation Company Limited, Hong Kong.

Jan. 2019

- Introduced Lisee technology in front of selection committee panel and won TSSSU 2019-20 funding of the amount of HK\$300,000.

IEEE Int. Conf. on CYBER Technology in Automation, Control, and Intelligent Systems [IEEE-CYBER 2018]

Tianjin, China

Presenter as author

Jul. 2018

- Introduced [PCR-Pro](#) for the estimation of information matrix and scale difference for pose graph SLAM.

IET-Young Professionals Exhibition and Competition (IET-YPEC) 2018

CityU, Hong Kong

Presenter as COO Lisee Technology Corporation Company Limited, Hong Kong.

Jul. 2018

- Introduced Lisee technology at IET-YPEC 2018.

Campus Challenge @ HKUST Transportation

HKUST, Hong Kong

Presenter as participant

May 2018

- HKUST got 80+ proposals for this competitions, and arranged a roadshow. My proposal got selected for the audience award after getting maximum votes.

High Tech Innovation and Entrepreneurship

HKUST, Hong Kong

Presenter as Entrepreneur

Dec. 2017

- Introduced my study related to the flying cars.

World First Youth Robotics Development Forum 2017

Shangri-La Shenzhen, China.

Presenter as author.

Nov. 2017

- Introduced my work related to multi-agent SLAM system.

Honors & Awards

July, 2021	Educational Technology Innovation: Gold Award , The Teaching and Learning Innovation Expo, CUHK	CUHK, Hong Kong
July, 2021	Pedagogical Innovation: Silver Award , The Teaching and Learning Innovation Expo, CUHK	CUHK, Hong Kong
July, 2018	Best Student Paper Finalist Award , IEEE Int. Conf. on CYBER Technology in Automation, Control, and Intelligent Systems	Tianjin, China
May, 2018	Audience Award , Campus Challenge - Transportation, HKUST	HKUST, HK
Nov, 2017	Best Mapper Award , Robotics and Multi-Perception Lab, HKUST	Shenzhen, China
May, 2017	Completion Award , The Graduate Teaching Assistant Training, HKUST	HKUST, HK
Dec, 2016	Best External Award , Robotics and Multi-Perception Lab, HKUST	Shenzhen, China
Nov, 2016	Completion Award , Author Workshop, Elsevier Publishing Campus	HKUST, HK
Sep 2016	CCMS PhD Scholarship , Won Chief Minister Marit Scholarship (CCMS) scholarship award by Punjab Government (PEEF), Pakistan for pursuing my PhD studies at HKUST, HK	Lahore, PK
Sep 2013	Participant , Painting for peace (Pakistan-India-Afghanistan) event, a part of the Harmony initiative	Islamabad, PK
Apr 2013	Participant , Case Spark in business plan competition	Islamabad, PK
Apr 2013	Participant , Case EPICA	Islamabad, PK
Apr 2013	Participant , How to be a Better Playwright Workshop by PNCA and US Embassy Pakistan	Islamabad, PK
Oct 2012	Participant , Punjab Youth Festival	Lahore, PK
Oct 2011	Participant , Int'l Youth Conference and Festival	Islamabad, PK
Sep 2011	Participant , Asian Youth Conference and Peace Festival	Lahore, PK
Apr 2011	Completion Award , Acting and Directing Theater Workshop by PNCA and US Embassy Pakistan	Islamabad, PK
Dec 2011	Participant , National Drama Festival	Islamabad, PK
Dec 2010	Instructor of short Course , PHP Web Development' at the Department of Electronics, QAU	Islamabad, PK
Apr 2013	Completion Award , Running River Rafting course by Adventure Foundation of Pakistan	Mansehra, PK
Apr 2013	Completion Award , Paragliding course by QAU Adventure Club	Islamabad, PK
Dec 2008	Award of Honor , Finance Secretary of Electronics Student Society (ESS) at QAU	Islamabad, PK

Scientific Review Services and Academic Services

Journal Referee Services

T-RO	Peer Reviewer , IEEE Transactions on Robotics	2026
RA-L	Peer Reviewer , IEEE Robotics and Automation Letter	2026, 25, 24, 23, 22
TJS	Peer Reviewer , The Journal of Supercomputing	2024
T-C	Peer Reviewer , IEEE Transactions on Cybernetics	2022

Conference Referee Services

IROS	Peer Reviewer , IEEE/RSJ RA-L and International Conference on Intelligent Robots and Systems	2024, 23, 21, 20, 18
ICRA	Peer Reviewer , IEEE/RSJ International Conference on Robotics and Automation	2024, 23, 22, 21, 20
UF, US	Poster Judge , UF/IFAS Agricultural and Biological Engineering Poster Symposium	2024

Teaching Assistant Positions

HKUST, HK	Teaching Assistant , EESM 5547 - Multimedia Signal Processing	S 20, S 19, S 18
HKUST, HK	Teaching Assistant , ELEC4820 - Medical Imaging	F 18
HKUST, HK	Teaching Assistant , ELEC 1010 - Electronic and Information Technology	S 17

Skills

Robotics and Computer Vision	Ubuntu, ROS2, PCL, Docker, Catkin, OpenCV, OpenGV, Ceres Solver, g2o, Libpointmatcher
Machine Learning / Deep Learning	Jupyter, Pytorch, Tensorflow, Keras, CUDA, Cluster (Hipergator)
Programming	Python, C/C++, Matlab, git, LaTeX, FPGA, Verilog, SLURM
Web	mkdocs, obsidian
Languages	English, Urdu

Research

G: [Google Scholar](#) | **ORCID iD:** [0000-0002-3512-4279](#)

Journal Publications

Validation and Estimation of Blueberry Yield Using Image-Based Object Detection, Maturity Classification, and Canopy Architecture Under Field Conditions	<i>Journal</i>
Paul Adunola and Tyler J. Schultz and Bruno Leme and M. Usman Maqbool Bhutta and Minz and M. Amman and Luis Felipe Ventorim Ferrão and Patricio Muñoz	Jul 2026
Under Review	
In-Sensor Visual Perception and Inference	<i>Journal</i>
Yanan Liu and Rui Fan and Jianglong Guo and Hepeng Ni and M. Usman Maqbool Bhutta	Jul 2023
2023 Intelligent Computing	
Robust mobile robot navigation in cluttered environments based on hybrid adaptive neuro-fuzzy inference and sensor fusion	<i>Journal (IF: 8.8+)</i>
Muhammad Husnain Haider, Zhonglai Wanga, Abdullah Aman Khan, Hub Ali, Hao Zheng, Shaban Usman, Rajesh Kumar, M. Usman Maqbool Bhutta, Pengpeng Zhi	Aug 2022
2022 Journal of King Saud University-Computer and Information Sciences.	
MAFNet: Segmentation of Road Potholes with Multi-modal Attention Fusion Network for Autonomous Vehicles	<i>Journal (IF: 5.3+)</i>
Zhen Feng, Yanning Guo, Qing Liang, M. Usman Maqbool Bhutta, Hengli Wang, Ming Liu, and Yuxiang Sun	Aug 2022
2022 IEEE Transactions on Instrumentation & Measurement	
Why-So-Deep: Towards Boosting Previously Trained Models for Visual Place Recognition	<i>Journal (IF: 5.4+)</i>
M. Usman Maqbool Bhutta*, Yuxiang Sun, Darwin Lau and Ming Liu	Jan 2022
IEEE Robotics and Automation Letters (RA-L)/ICRA, 2022	
Loop-box: Multi-Agent Direct SLAM Triggered by Single Loop Closure for Large Scale Mapping	<i>Journal (IF: 11.7+)</i>
M. Usman Maqbool Bhutta*, Manohar Kuse, Rui Fan, Yanan Liu and Ming Liu	Oct 2020
2022 IEEE Transactions on Cybernetics	
Efficient Reset Design Techniques	<i>Journal Paper</i>
Qamar, Zoha, Bushra Yaseen, and Usman Maqbool	2015
Published in Science International Lahore	

Conference Proceedings

CaMuViD: Calibration-Free Multi-View Detection	<i>Conference Paper</i>
Amir Etefaghi Daryani, M. Usman Maqbool Bhutta, Byron Hernandez, Henry Medeiros	June 2025
Conference on Computer Vision and Pattern Recognition (CVPR), 2025	
Autonomous Mobile Robot Navigation using Adaptive Neuro Fuzzy Inference System	<i>Conference Paper</i>
Muhammad Husnain Haider and Hub Ali and Abdullah Aman Khan and Hao Zheng and M. Usman Maqbool Bhutta and Shaban Usman and Pengpeng Zhi and Zhonglai Wang	May 2022
2022 International Conference On Innovations And Development Of Information Technologies And Robotics (IDITR), 2022	
Scale-Adaptive Pothole Detection and Tracking from 3-D Road Point Clouds	<i>Conference Paper</i>
Wu, Rigen and Fan, Jiahe and Guo, Libo and Qiao, Lei and Bhutta, M. Usman Maqbool and Hosking, Brett and Vityazev, Sergey and Fan, Rui	Aug 2021
2021 IEEE International Conference on Imaging Systems and Techniques (IST), 2021	

Smart-Inspect: Micro Scale Localization and Classification of Smart Phone Glass Defects for Industrial Automation

M Usman Maqbool Bhutta*, Shoaib Aslam, Peng Yun, Jianhao JIAO and Ming Liu

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2020)

Conference Paper

Oct 2020

Multiple Lane Detection Algorithm Based on Optimised Dense Disparity Map Estimation

Han Ma, Yixin Ma, Jianhao Jiao, M. Usman Maqbool Bhutta, Mohammud Junaid Bocus, Lujia Wang, Ming Liu and Rui Fan

Accepted in IEEE International Conference on Imaging Systems and Techniques (IST 2018)

Conference Paper

Dec 2018

PCR-Pro: 3D Sparse and Different Scale Point Clouds Registration and Robust Estimation of Information Matrix For Pose Graph SLAM

M. Usman Maqbool Bhutta* and Ming Liu

Presented in IEEE Int. Conf. on CYBER Technology in Automation, Control, and Intelligent Systems (IEEE-CYBER 2018), and won **Best Student Paper Finalist Award**.

Conference Paper

Jul 2018

An intelligent approach for robust detection and recognition of multiple color and font styles automobiles license plates: A feature-based algorithm

M. Usman Maqbool Bhutta*, Hasan Mahmood, and Haroon Malik

Published in IEEE International Conference on Audio, Language and Image Processing (ICALIP 2014).

Conference Paper

Jan 2015

Patent

Semi-Supervised Learning Approach for Smart Phone Glasses Defect Inspection

M. Usman Maqbool Bhutta, Jianhao Jiao, Peng Yun, Yuan Wang, Xuebin Sun, Ming Liu

U.S. Provisional Patent Application No. 62/974,937, Filing Date 01/06/2020.

US Patent

Jan 2020

Automatic Surface Defect Detection System for Smart Phone Glass Based on Deep Learning

Yuan Wang, Jianhao Jiao, Peng Yun, M. Usman Maqbool Bhutta, Xuebin Sun, Ming Liu

U.S. Provisional Patent Application No. 62/974,938, Filing Date 01/06/2020.

US Patent

Jan 2020

References

Dr. Shoaib Ehsan

Associate Editor · IEEE Transactions on Robotics (T-RO) & IEEE Robotics and Automation Letters (RA-L)

Editor / Academic Contact

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PhD Co-Supervisor

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Postdoc Supervisor

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